12 th HACKER JUGAAD NOTES

MATRICES

Representation to Matrix:

$A = [a_{ij}]_{m \times n}$

i^{th} row, j^{th} column, m×n → order

③ × ② → row × column

eg. $A = [a_{ij}]_{3 \times 2}$ (R = 3, C = 2)

$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$ → 5

eg. if $a_{ij} = \frac{i^2 + j}{2}$

then $a_{31} = \frac{3^2 + 1}{2} = 5$

$i = 3$

$j = 1$

Order of Matrix:

m × n

TYPES OF MATRICES

Row Matrix → A single row matrix is called a row matrix or a row vector. i.e a matrix $A = [a_{ij}]_{m \times n}$ is said to be row matrix, if m=1

Column Matrix → A single column matrix is called a column matrix or a column vector. i.e - a matrix $A = [a_{ij}]_{m \times n}$ is said to be column matrix if n=1

Null or Zero Matrix → If all the elements of matrix are equal to zero, then it is called a null matrix and is denoted by $O_{m \times n}$ or O.

SQUARE MATRICES

Number of rows = No. of Columns

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{33} & a_{34} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}_{n \times n}$$

→ Order = n × n OR n

→ Total Elements = n^2

→ Conjugate Elements

→ Principle / Leading diagonal ⌒ $j = i$

→ Number of elements in upper triangle ↘ $\frac{n^2 - n}{2}$

→ Number of elements in lower triangle ↗ $\frac{n^2 - n}{2}$ (equal)

Conjugate elements

a_{ij} & a_{ji}

eg. a_{23} & a_{32} — conjugate

$A_{3 \times 3} =$

$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

a_{12}, a_{13}, a_{23} → U.T.P. $(i < j)$

a_{11}, a_{22}, a_{33} → a_{ij} $(i = j)$

a_{31}, a_{32} → L.T.P $(i > j)$

TYPES OF SQUARE MATRICES

Scalar Matrix → A square matrix in which all the diagonal element are equal and all other elements equal to zero is called a Scalar matrix. i.e. in a scalar matrix $a_{ij} = K$ for $i = j$ and $ij = 0$ for $i \neq j$.

Unit Matrix or Identify Matrix → A square matrix in which all its diagonal elements are equal to 1 and all other elements equal to zero is called a unit matrix or identify matrix.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

OPERATIONS ON MATRICES

1. ADDITION & SUBTRACTION

Note : Conformable for addition / Substraction : Order is same.

2. MULTIPLICATION BY SCALAR

Matrix ke sabhi elements multiply honge !!

3. EQUALITY OF TWO MATRICES

Two matrices A and B are said to be equal if :

(i) They have same order.

(ii) $a_{ij} = b_{ij}$ $\forall$ i, j

↳ Position wise elem. भी equal हों

Examples : 1. $\begin{bmatrix} -1 \\ 5 \\ 6 \end{bmatrix}_{3\times1} + \begin{bmatrix} 2 \\ -2 \\ 7 \end{bmatrix}_{3\times1} + \begin{bmatrix} 1 \\ 3 \\ 13 \end{bmatrix}_{3\times1}$

2. $A = \begin{bmatrix} 3 & -1 \\ 2 & 0 \\ 4 & -3 \end{bmatrix}$

↳ $5A = \begin{bmatrix} 15 & -5 \\ 10 & 0 \\ 20 & -15 \end{bmatrix}$

3. $\begin{bmatrix} a & b \\ x+y & x-y \end{bmatrix}_{2\times2} = \begin{bmatrix} -2 & a \\ 3 & 5 \end{bmatrix}_{2\times2}$

$a = -2 \quad b = a = -2 \quad x + y = 3$

$x - y = 5$

MATRIX MULTIPLICATION

Multiplication of Matrices → Product of two matrices exists only if number of column of first matrix is equal to the number of rows of the second. Let A be m x n and B be n x p matrices. Then the product of matrices A and B denoted by A.B is the matrix of order m x p whose (i.j) the elements is obtained by adding the products of corresponding elements of ith row of A and jth column B.

When ?? $A_{m\times n}$ $B_{p\times q}$ $= C$

$A_{3\times5}$ $B_{5\times2}$ $= C$

$n = p$

How ?? Row x Column

पहली वाली Matrix की Row × दूसरे Matrix के Column

TRANSPOSE OF MATRIX

$$A = \begin{bmatrix} 3 & -1 \\ 2 & 0 \\ 4 & 5 \end{bmatrix}_{3\times2}$$

$A_{m\times n}$

Transpose of matrix $A = \underbrace{A^T / A'}_{n \times m}$

2×3

$$= \begin{bmatrix} 3 & 2 & 4 \\ -1 & 0 & 5 \end{bmatrix}$$

Properties :

1. $(A')' = A$
2. $(KA)' = K(A)'$ (K ↓ number)
3. $(A \pm B)' = A' \pm B'$
4. Reversal Law :

$$(AB)' = B'A'$$

SYMMETRIC & SKEW SYMMETRIC MATRIX

SYMMETRIC MATRIX : A square matrix $A = [a_{ij}]$ is said to be a symmetric matrix if $a_{ij} = a_{ji}$ for all i and j.

$$A^T = A$$

SKEW SYMMETRIC MATRIX : A square matrices $A = [a_{ij}]$ is said to be a skew symmetric matrix if $a_{ij} = -a_{ji}$ for all i and j.

Symmetric

Visually $A = \begin{bmatrix} a & x \\ x & b \end{bmatrix}$, $\begin{bmatrix} (2) & (3) \\ (3) & -1 \end{bmatrix}$ (a_{11}, a_{12}, a_{21})

Symm Matrix $A = \begin{bmatrix} a & p & q \\ p & b & x \\ q & x & c \end{bmatrix}$ $\begin{bmatrix} 3 & -2 & 0 \\ -2 & -1 & 7 \\ 0 & 7 & 5 \end{bmatrix}$

Skew Symmetric

$A = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$

$A = \begin{bmatrix} 0 & 2 & 5 \\ -2 & 0 & -6 \\ -5 & 6 & 0 \end{bmatrix}$

Theoritical $A^T = -A$

INVERSE OF MATRIX

$AB = BA = I$

A & B are called as Inverse of each other.

Inv. of matrix $A = A^{-1}$

$A^{-1}A = I = AA^{-1}$

ELEMENTARY OPERATIONS

The operations known as elementary row operation on a matrix are —

(i) The interchange of any two rows of a matrix.

(ii) The multiplication of every element of a row by a non-zero element (constant).

(iii) The addition of the elements of a row, the product of the corresponding elements of any other row by any non-zero constant.

① $R_i \leftrightarrow R_j$ or $C_i \leftrightarrow C_j$

② $R_i \longrightarrow KR_i$ or $C_i \longrightarrow KC_j$

③ $R_i \longrightarrow R_i + KR_j$ or $C_i \longrightarrow C_i + KC_j$

e.g. $A = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$ $R_1 \leftrightarrow R_2 \Rightarrow \begin{bmatrix} 3 & 5 \\ 2 & -1 \end{bmatrix}$

elementary row op $R_2 \rightarrow -3R_2 \Rightarrow \begin{bmatrix} 2 & -1 \\ -9 & -15 \end{bmatrix}$

$R_2 \rightarrow R_2 + 5R_1 \Rightarrow \begin{bmatrix} 2 & -1 \\ 13 & 0 \end{bmatrix}$